

Physics (042)



SECTION - A

(B) Becomes greater than C

(A) $\frac{\alpha}{r}$

(D) $\frac{4R}{3}$

(B) 5 cm

(C) 0.196 Am²

(D) 69 V

(A) Infrared rays

(B) [M⁰L²T⁻²]

(A) x-rays

$$\frac{A\epsilon_0}{\frac{x + d \cdot x}{K} \cdot \infty} = \frac{A\epsilon_0 K}{x, d > x}$$

$$C = \frac{A\epsilon_0}{d}$$

NIA.

$$\frac{10 \times \pi \times (14)^2}{\pi} \cdot |E| = \frac{d \cdot \hat{r}}{r^2}$$

$$dE = -\frac{dV}{dx}$$

$$10 \times 196$$

$$106 \times 100$$

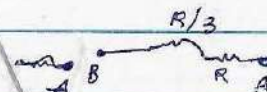
=

$$-\int \frac{dV}{r^2} = \left(\frac{d}{r} \right) \infty$$

$$= 0 - \frac{d}{x}$$

$$16t + 5$$

$$64 + 5$$



$$q \cdot 10^4 = 1 \cdot m \cdot 10^{12}$$

$$c^2 = \frac{1}{\mu \epsilon}$$

$$\frac{q}{m} = \frac{1 \times 10^8}{2}$$

$$L\bar{T}^{-1}$$

$$r = \frac{mv}{qB} = \frac{2 \times 10^{-8} \times 10}{4 \times 10^{-6}}$$

$$\frac{mv}{\lambda h}$$

$$= 0.5 \times 10^{-1}$$

$$= 0.05 \text{ m}$$

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Question Number



(A) f_0 and f_e small, and $f_e > f_0$

$2a.$

$0, 4a^2.$

(B) 0 and $4a^2$

$$\sqrt{2mKE} = \frac{h}{\lambda}$$

$$\lambda \alpha = \frac{h}{\sqrt{m_p(KE)} p.}$$

$$\lambda p. \sqrt{m_\alpha(KE)} \alpha.$$

(C) $\frac{1}{4}$

$$= \sqrt{\frac{1}{4}} \cdot \sqrt{\frac{1}{4}} = \frac{1}{4}.$$

13. (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

14. (C) Assertion (A) is true, but Reason (R) is false.

15. (D) Both Assertion (A) and Reason (R) are false.

16. (C) Assertion (A) is true, but Reason (R) is false.



17.

SECTION - B

ν_0 = threshold frequency = 3.6×10^{14} Hz.

ν = frequency of incident light = 6.8×10^{14} Hz

By photoelectric equation,

$$eV_0 = h\nu - h\nu_0$$

where $V_0 \rightarrow$ cut off potential of photoelectron.

$$(1.6 \times 10^{-19}) V_0 = 6.63 \times 10^{-34} (6.8 \times 10^{14} - 3.6 \times 10^{14})$$

$$= 6.63 \times 10^{-34} (3.2 \times 10^{14})$$

$$V_0 = \frac{6.63 \times 10^{-34} (3.2 \times 10^{14})}{(1.6 \times 10^{-19})}$$

$$= 6.63 \times 10^{-34} (2 \times 10^{-5})$$

$$= 13.26 \times 10^{-39} \text{ V}$$

$$= 1.326 \times 10^{-38} \text{ V} //$$

18. (b)

For Eng² two coherent sources each of intensity I ,
intensity of central maxima = $4I = I_0$.

[at central maxima, phase difference (ϕ) = 0
Intensity = $4I \cos^2(\phi/2) = 4I$]

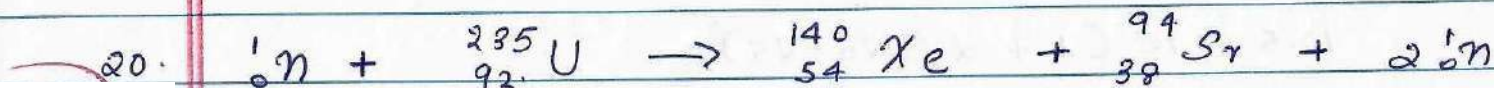
$$\frac{\Delta x}{\lambda} = \frac{\phi}{2\pi}$$

$$(\Delta x = \frac{\lambda}{3})$$

$$\frac{\lambda}{3\lambda} = \frac{\phi}{2\pi} \Rightarrow \phi = \frac{2\pi}{3}$$

$$\text{Intensity} = 4I \cos^2(\phi/2) = 4I \cos^2(\pi/3) = \frac{4I}{4} = I$$

$$= \frac{I_0}{4} //$$



$$\Delta m = m[({}^{235}_{92}\text{U}) + ({}^1_0n)] - [m({}^{140}_{54}\text{Xe}) + m({}^{94}_{38}\text{Sr}) +$$



$$2m({}_0^1n_a)]$$

$$\Delta m = 235.04393u + 1.00866u - (139.92164 + 93.91536) - 2(1.00866)$$

$$= 235.04393 - 1.00866 - (233.837)$$

$$= 234.03527 - 233.837 = 0.199827u$$

$$BE = \text{Binding energy} = \Delta mc^2$$

$$= 0.199827 \times 931 \text{ MeV}$$

$$= 184.18937 \text{ MeV}$$

Q1.

Resistance of wire at $25^\circ\text{C} = R_{25} = 10\Omega$

Resistance of wire at $125^\circ\text{C} = R_{125} = 10.5\Omega$

(i)

$$R_{125} = R_{25}(1 + \alpha \Delta T)$$

$$10.5 = 10(1 + \alpha(125 - 25))$$

$$10.5 = 10 + 10(\alpha)(100)$$

$$0.5 = 1000\alpha$$

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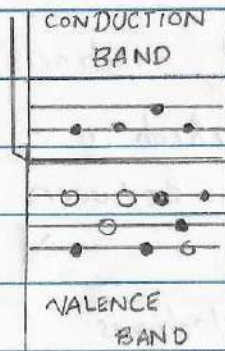


$$\alpha = 0.5 \times 10^{-3} \text{ } ^\circ\text{C}^{-1} = 5 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$$

$$\begin{aligned} (90) \quad R_{425} &= R_{25} (1 + \alpha \Delta T) \\ &= 10 \left(1 + 5 \times 10^{-4} (425 - 25) \right) \\ &= 10 + 10 \times 5 \times 10^{-4} (4) \times 10^2 \\ &= 10 + 5 \times 4 \times 10^{-1} \\ &= 10 + 20 \times 10^{-1} = 10 + 2 \\ &= 12 \Omega // \end{aligned}$$

SECTION-C.

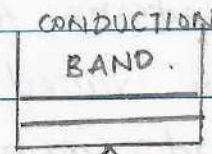
92. (a)



Conductors

○ → holes

• → electrons



VALENCE BAND

Insulators

$E_g \rightarrow$ energy gap.

$$139.92164.$$

$$\begin{array}{r} 93.91536 \\ 100 \end{array}$$

$$233.83700$$

$$295.04392.$$

$$- 1.00866.$$

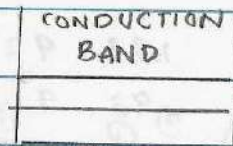
$$234.03527$$

$$233.83700$$

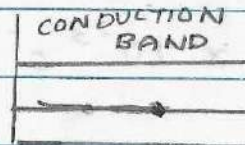
$$.79827$$

$$\times 931$$

$$\begin{array}{r} 1.79827 \\ 53904810 \\ 100 \\ 16844300 \\ 224.8937 \end{array}$$



Semiconductors.

at $T = 0K$.

Semiconductor.

at $T > 0K$.

→ At $T = 0K$, semiconductor acts as an insulator due to absence of free electrons for conduction.

→ When temperature is raised (room temperature), electrons in the valence band gain thermal energy, which is sufficient to cross over the small energy gap between valence band and conduction band.

→ This when ~~valence band~~ ~~electrons~~ ~~vacancy~~ ~~of~~ ~~electrons~~ produced in the valence band generate holes.

→ This is how electron-hole pairs are generated at room temperature in semiconductors.



23.

- (b) → Carbon and silicon, both contain 4 valence electrons in second shell and third shell respectively.
- As valence electrons in carbon are much closer to the nucleus, compared to that of silicon, they are strongly held by the nucleus.
- Energy required to remove an electron i.e Ionisation Enthalpy, of carbon is much greater than silicon.
- Hence, due to unavailability of free electrons, carbon acts as an insulator. Whereas, silicon acts as a semiconductor.

Eg carbon > Eg silicon Eg → energy gap.

23.

Generally, $C = \frac{A \epsilon_0 k}{d}$

A → Area of parallel plate

d → separation between parallel plate

(a) $C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$

(in series)

where

$$C_1 = \frac{k A \epsilon_0}{d/2} = kC$$

&

$$C_2 = \frac{A \epsilon_0}{d} = C$$



$$C_{eq} = \frac{kC}{kC + C} = \frac{C^2 (K)}{C (K+1)} = \frac{KC}{K+1}$$

$$= \left(\frac{K}{K+1} \right) \frac{2A\epsilon_0}{d} \text{ Farads} //$$

(b) $C_{eq} = C_1 + C_2$ (parallel connection).

$$C_1 = \frac{A}{2} \frac{\epsilon_0 k}{d} = kC, \quad C_2 = \frac{A}{2} \frac{\epsilon_0}{d} = C$$

$$C_{eq} = kC + C = C(K+1) \\ = \left(\frac{K+1}{2} \right) \frac{A\epsilon_0}{d} \text{ Farads} //$$

24. $d = 1\text{mm} = 1 \times 10^{-3} \text{m}$

$D = 1\text{m}$

$\lambda_1 = 500\text{nm} = 5 \times 10^{-7} \text{m}$

$\lambda_2 = 600\text{nm} = 6 \times 10^{-7} \text{m}$

(a) Position of First maxima of $\lambda_1 = \frac{\lambda_1 D}{d}$ ($\Delta x = \lambda_1$)
(x_1)



Position of ~~see~~ first maxima for $\lambda_2 = \frac{\lambda_2 D}{d}$ ($\Delta x = (1) \lambda_2$,
(λ_2)

$$\begin{aligned}\text{Distance between first maxima} &= x_2 - x_1 \\ &= \frac{D}{d} (\lambda_2 - \lambda_1) \\ &= \frac{1}{1 \times 10^{-3}} (6 - 5) \times 10^{-7} \text{ m} \\ &= 10^{-4} \text{ m} //\end{aligned}$$

(b) Let ^{position of} n_1 th bright of $\lambda_1 = \frac{n_1 \lambda_1 D}{d}$

Let position of n_2 th bright of $\lambda_2 = \frac{n_2 \lambda_2 D}{d}$

For both to coincide,

$$n_1 \frac{\lambda_1 D}{d} = n_2 \frac{\lambda_2 D}{d}$$

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{600}{500} = \frac{6}{5}$$

$\therefore$ 6th bright fringe of λ_1 coincides with 5th bright fringe

11
22. (first)

Minimum distance from central maxima

$$= \frac{6 \lambda D}{d} = \frac{6 \times 5 \times 10^{-7} \times 1}{10^{-3}}$$

$$= 30 \times 10^{-4} \text{ m}$$

$$= 3 \times 10^{-3} \text{ m} = 3 \text{ mm}.$$

25. Half wave Rectifier.

⇒ Rectifier where only ~~1~~ one half cycle of ~~output~~ voltage is rectified

Full wave Rectifier

⇒ Rectifier where both cycles of ^{output} ~~input~~ voltage is rectified.

HALFWAVE RECTIFIER

⇒ only one ^{half} cycle of output voltage is rectified

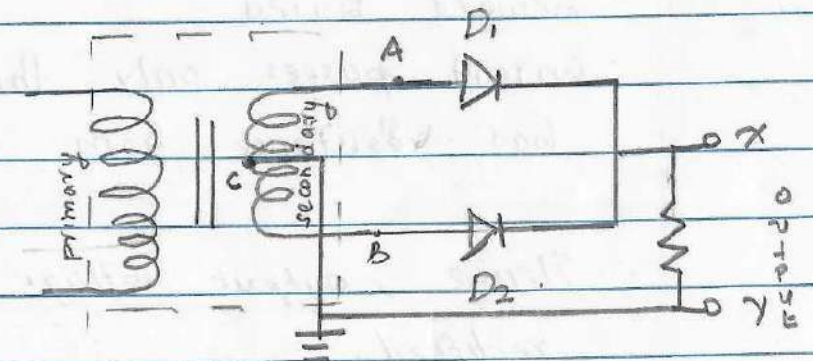
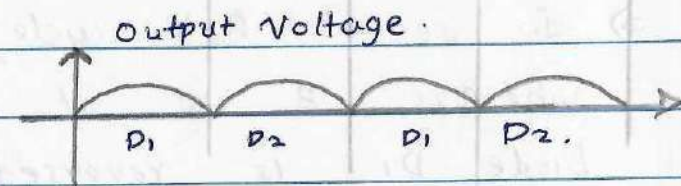
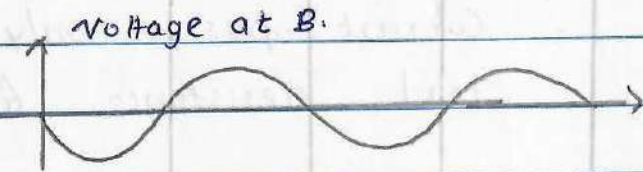
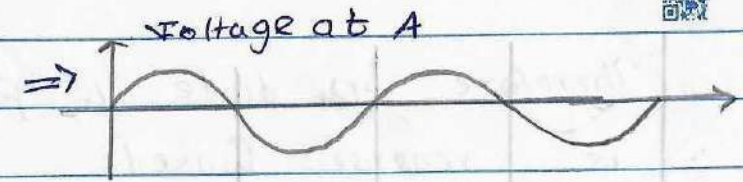
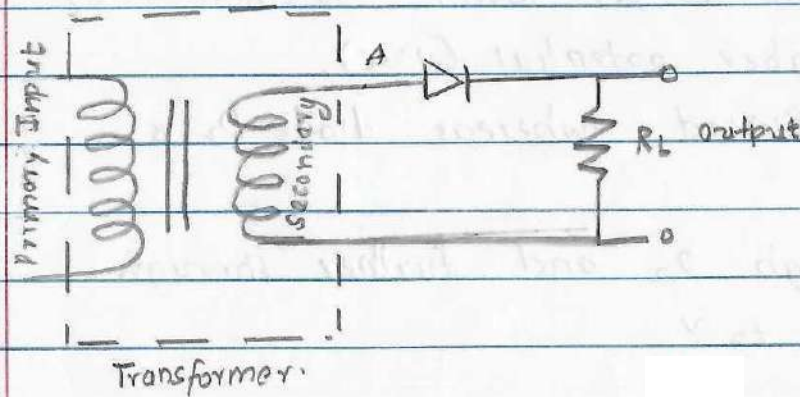
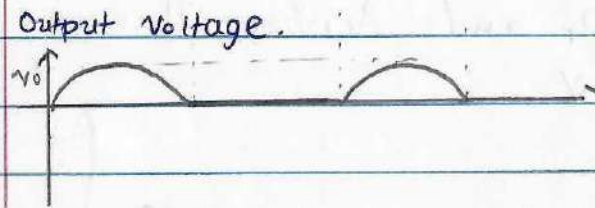
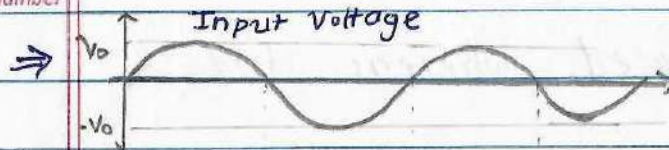
⇒ Makes use of only one diode

FULL WAVE RECTIFIER.

→ Both half cycles of output voltage are rectified

⇒ Makes use of two four diodes.

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Question Number



⇒ Centre tapping is not provided ⇒ Centre tapping is provided

* During first positive half cycle in full wave rectifier, ~~point~~ A is at higher (+ve) potential while B is in lower potential.



Therefore, D_1 diode is forward Biased whereas diode D_2 is reverse Biased.

$\therefore$ Current passes only through D_1 and further through load resistance from X to Y .

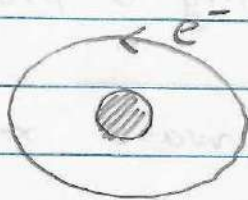
$\Rightarrow$ In second half cycle, A is at lower potential ($-$) whereas B is at higher potential ($+ve$).

Diode D_1 is reversed Biased whereas diode D_2 is forward Biased.

$\therefore$ Current passes only through D_2 and further through load resistance from X to Y .

$\therefore$ Hence, output voltage across load resistance is fully rectified.

26. (a)



Magnetic dipole moment = NIA

① here, Number of turns (N) = 1

② current = $I = \frac{q}{t} = \frac{e}{t}$



$$t = \frac{2\pi r}{v} \quad \text{where } v \text{ is velocity of } e^-.$$

$r \Rightarrow$ radius of loop / orbit.

$$\therefore I = \frac{ev}{2\pi r} //$$

$$\odot A = \text{Area of loop} = \pi r^2.$$



$$\mu = NIA.$$

$$\therefore \mu = 1 \left(\frac{-ev}{2\pi r} \right) \pi r^2 = -\frac{evr}{2} //$$

Direction of dipole moment is out of the plane. $\odot$
(By right hand thumb rule).

$$(b). \quad \vec{\mu} = -\frac{evr}{2}.$$

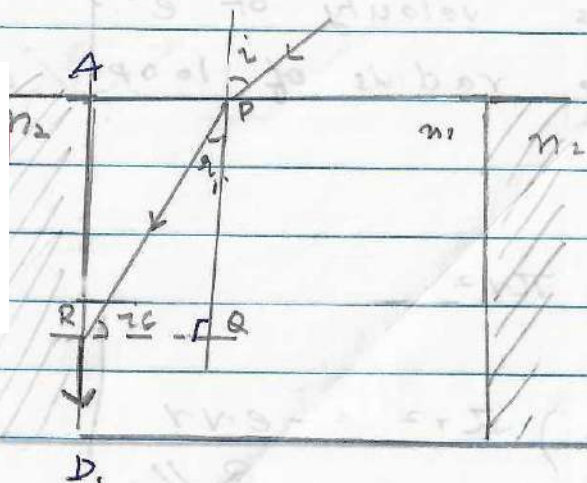
$$\vec{L} = \text{angular momentum} = mvr.$$

$$\therefore vr = \frac{\vec{L}}{m}.$$

$$\vec{\mu} = -\left(\frac{e}{2}\right) \frac{\vec{L}}{m} = \left(\frac{-e}{2m}\right) \vec{L} //$$



27.



$$n_2 = 1.25,$$

If light grazes out at AD,

$$\angle x_2 = 90^\circ$$

$\Rightarrow$ By Snell's Law, at AD

$$n_1 \sin i_1 = n_2 \sin i_2$$

$$\sin i = \frac{n_2}{n_1} \quad (i = 90^\circ)$$

$$i_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) = i_c$$

∴ it is the critical angle when light travels from glass to liquid.

From triangle PQR,

$$g_1 + i0 = q_0$$

$$\mu_c = 90 - 2c$$

$$\sin \alpha_1 = \frac{\sin}{\cos} (90 - i_c) = \sin / \cos i_c$$

$$= \sqrt{1 - \sin^2 i}$$

$$= \sqrt{1 - \left(\frac{1.25}{1.5}\right)^2}$$



$$\sin \theta_1 = \sqrt{1 - \left(\frac{5}{6}\right)^2} = \sqrt{\frac{11}{6^2}} = \frac{\sqrt{11}}{6}$$

5x5
25x6

Applying Snell's law at AB.

$$1 (\sin i) = n_2 (\sin \theta_1)$$

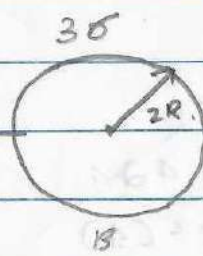
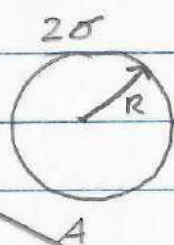
$$\sin i = 1.5 \frac{\sqrt{11}}{6} = \frac{3 \times \sqrt{11}}{2 \times 6} = \frac{\sqrt{11}}{4}$$

$$i = \sin^{-1}\left(\frac{\sqrt{11}}{4}\right)$$

(initially)

28

(a)



$$Q_A = 2\sigma (4\pi R^2)$$

$$Q_B = 3\sigma (4\pi (2R)^2)$$

$$= 3\sigma (4\pi R^2) \times 4$$

$$= 6 [2\sigma (4\pi R^2)] = 6 Q_A$$

On connecting,

$$V_A = V_B$$

$$\frac{k (Q_A + q)}{R} = \frac{k (Q_B + q)}{2R}$$

$$2 Q_A + 2q = Q_B + q$$

$$q = Q_B - 2 Q_A$$

$$= 6 Q_A - 2 Q_A$$

$$= 4 Q_A$$



$$\therefore \text{Charge density on A} = \frac{Q_A + q}{4\pi R^2}$$

$$= \frac{Q_A + 4Q_A}{4\pi R^2}$$

$$= \frac{5Q_A}{4\pi R^2}$$

$$= 5(2\sigma) = 10\sigma //$$

$$\therefore \text{Charge density on B} = \frac{Q_B - q}{4\pi R^2}$$

$$= \frac{6Q_A - 4Q_A}{4\pi R^2(4)}$$

$$= \frac{2Q_A}{4\pi R^2(4)} = \frac{2(2\sigma)4\pi R^2}{4\pi R^2(4)}$$

$$= \sigma //$$

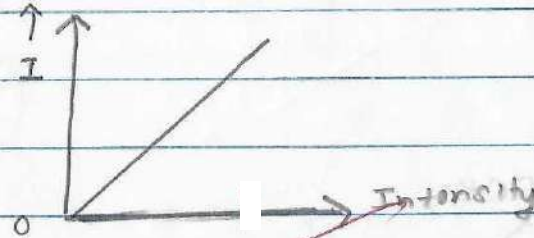
SECTION - D.

29. (i) (B) NBA
K.

(ii) (A) 0.25Ω

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Question Number(11) (B) 0.24×10^{-2} (iv) (B) 1.8×10^{-4} nm

30 (i) (C)



(ii) (D) remains same

(iii) (C) cutoff potential versus Frequency of incident light

(iv) (A) Caesium

SECTION - E

31 (a) (t)

$$S(0.2) = S(4.8)$$

$$S = \frac{6(2)}{48} = \frac{12}{48} = \frac{1}{4}$$

$$S(1/4) = \frac{3}{2}$$

$$6 + 1/4 = \frac{25}{4}$$

$$= \frac{3}{2} \times 2$$

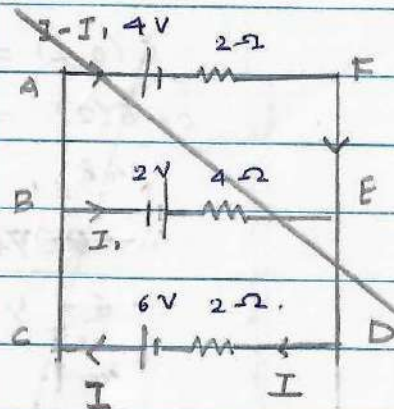
$$= \frac{25}{4} = 6.25$$

$$100 \times 6.25 = 625$$

$$\tau = NBIA$$

$$= 100 \times \frac{2}{10} \times 5 \times 10^{-3} \times 18 \times 10^{-4}$$

$$= 18 \times 10^{-5}$$



Using

 $\Rightarrow$ Kirchhoff's Loop rule states that

the sum of potential difference across a closed loop is zero.

Using KVL in loop BEDCB,

$$+2 - 4I_1 - 2I + 6 = 0 //$$

$$8 = 4I_1 + 2I //$$

$$4 = 2I_1 + I //$$

$$I = 4 - 2I_1 //$$

Using KVL in loop AFDCA,

$$-4 - 2(I - I_1) - 2I + 6 = 0 //$$

$$6 = 4 + 2I - 2I_1 + 2I //$$

$$2 = 4 + 4I - 2I_1 //$$

$$3 = 2 + 2I - I_1 //$$

$$3 = 2 + 8 - 4I_1 - I_1 //$$

$$5I_1 = 10 - 3 = 7 //$$

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Question Number



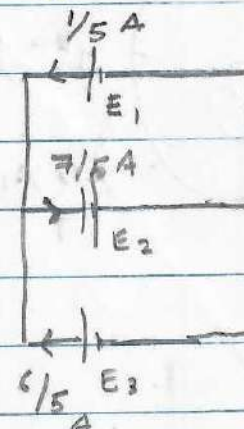
$$I_1 = 7/5$$

$$I = 4 - 2(I_1) = 4 - \frac{14}{5} = \frac{20-14}{5} = \frac{6}{5} \text{ A}$$

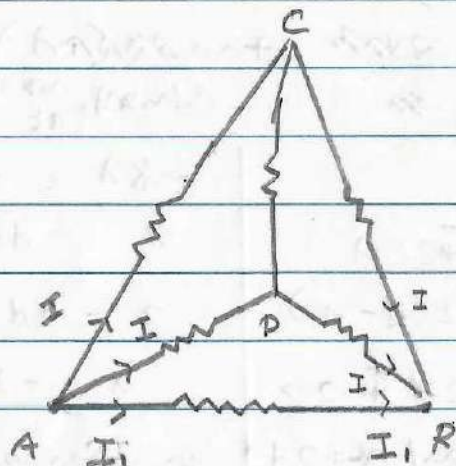
$$\text{Current through } E_1 = |I - I_1| = \frac{1}{5} \text{ A}$$

$$\text{Current through } E_2 = I_1 = \frac{7}{5} \text{ A}$$

$$\text{Current through } E_3 = I = \frac{6}{5} \text{ A}$$



(11)

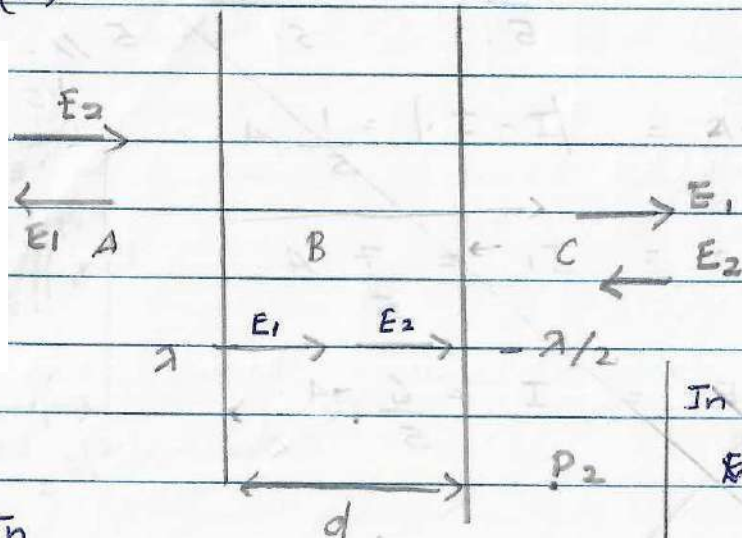




SECTION-C.

2830

(B)



E due to infinitely long wire

$$= \frac{\lambda}{2\pi\epsilon_0 r}$$

$$= \frac{2k\lambda}{r}$$

In

Region A, C //

$$\frac{2k\lambda}{d+x} = \frac{2k(\frac{\lambda}{2})}{x}$$

$$d+x = 2x$$

$$d = x //$$

at region C,

a point at distance $2d$ from
wire 1 has zero electric
field.

In region B, & A

$$E_p = \frac{2k\lambda}{x} - \frac{2k\lambda}{2(d-x)}$$

at B

at A

$$E_p = 0 //$$

$$\frac{2k\lambda}{x} = \frac{2k\lambda}{2(d-x)}$$

$$x = -2d + 2x$$

$$x = 2d$$

$$x = +2d$$

$$d-x = -ve \ x$$

There is no such point at A & B.
There is no such point at A & B.

$$\frac{2k\lambda}{x} = \frac{2k(\frac{\lambda}{2})}{d+x}$$

$$x = 2d + 2x$$

$$x = -2d$$

in region B,

From extra distance.

2d/3 from wire 1



32. (B) (i) When current passing through neighbouring coil is varied, flux linked with to coil₁ due to varying current in neighbouring coil₂ changes with time. This induces an ~~emf~~ emf in coil A. This phenomenon is called mutual induction.

Let current pass through neighbouring coil be I_2 , and flux linked with coil₁ due to neighbouring coil be Φ .

$$\text{Induced emf in coil A} = \mathcal{E} = - \frac{d\Phi}{dt} = - \frac{dI_2}{dt}$$

$$\mathcal{E} = - M \frac{dI_2}{dt}$$

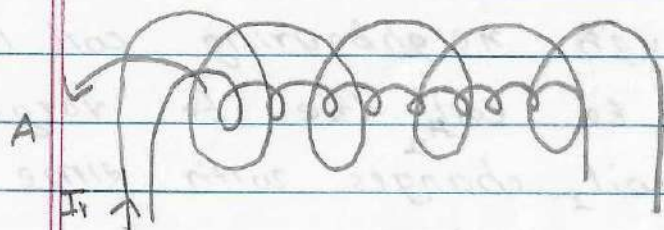
The constant of proportionality is called mutual inductance where M_{21} is the mutual inductance of coil 1 with respect to coil 2.

$$N_1 \Phi_1 \propto I_2$$

$N_1 \rightarrow$ no of turns in coil 1.

$$N_1 \Phi_1 = M_{12} I_2$$

Constant of proportionality is called mutual inductance.



Let

For large solenoid,

$$\text{current} = I_1.$$

$$(A_1) \text{ Area} = \pi r_1^2.$$

$$\text{Length} = l.$$

$$\text{number of turns per unit length} = n_1$$

For small solenoid,

$$(A_2) \text{ Area} = \pi r_2^2.$$

$$\text{Length} = l.$$

$$\text{number of turns per unit length} = n_2.$$

We know

$$N_2 \Phi_2 = M_{21} I_1.$$

$$N_2 (B A_2) = M_{21} I_1.$$

Assuming magnetic field to be constant along the centre,

$$B_1 = \mu_0 n_1 I_1.$$

$$\text{or } N_2 = n_2 l.$$

$$n_2 l (\mu_0 n_1 I_1) \pi (r_2)^2 = M_{21} I_1.$$



$$M_{21} = \mu_0 n_1 n_2 l \pi (r_2)^2$$

(11)

$$5 \text{ mV} = \left| L \frac{dI}{dt} \right|$$

$$5 \times 10^{-3} = L \left(\frac{2}{40} \right)$$

$$L = 20 \times 5 \times 10^{-3} \text{ H} = 10^{-1} \text{ H} //$$

$$\Phi = L I$$

$$= 10^{-1} \times \frac{1}{2} \text{ WB.}$$

$$= 0.5 \times 10^{-1} \text{ WB}$$

$$= 5 \times 10^{-2} \text{ WB.}$$

$$\frac{I}{10} = \frac{2 \times 10}{40} = \frac{2}{4} = \frac{1}{2}$$

$$I_t = \frac{2}{40} \text{ t.}$$

33. (b)

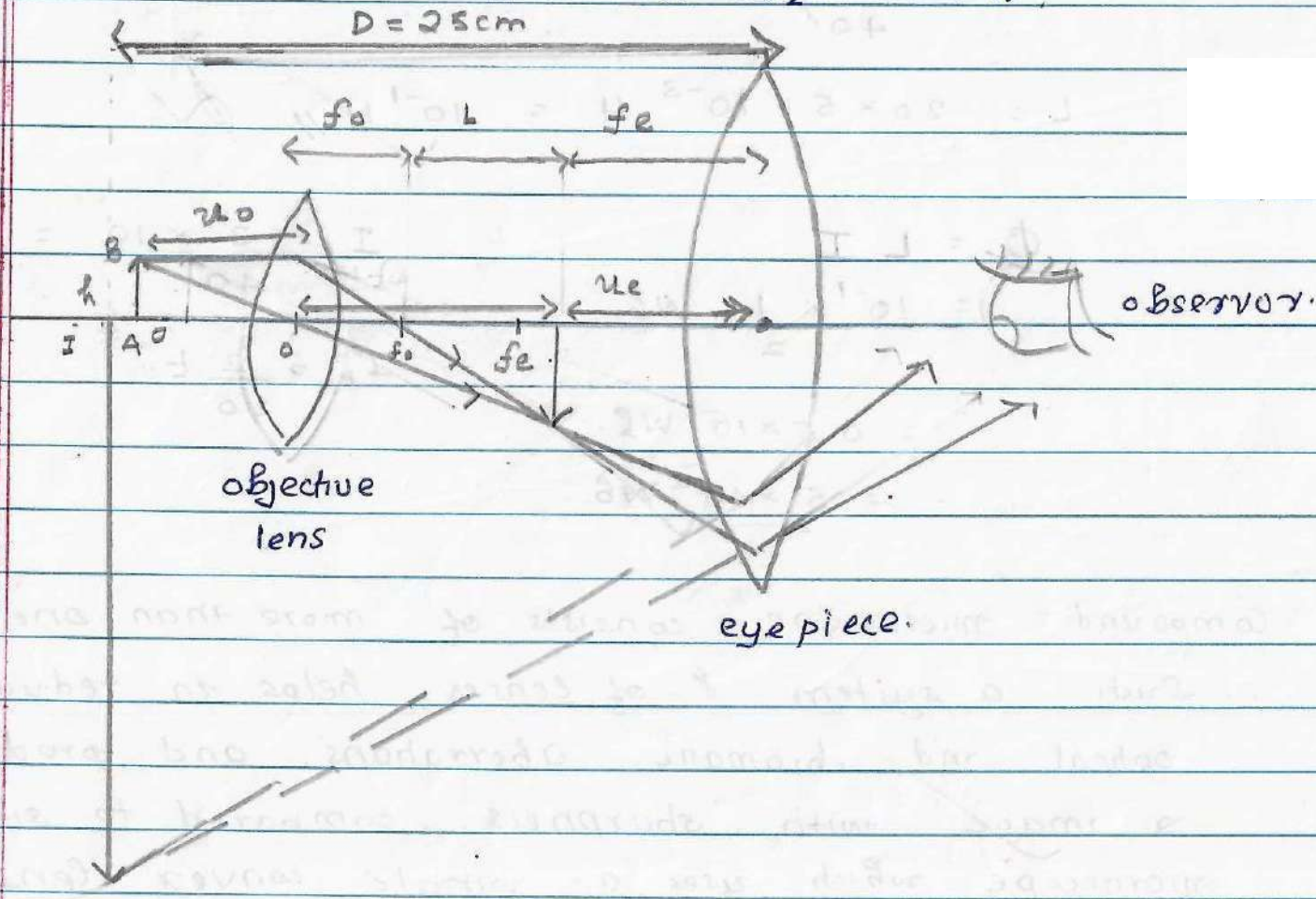
→ Compound microscope consists of more than one lens.

∴ Such a system of lenses, helps in reducing optical and chromatic aberrations, and produce a image with sharpness, compared to simple microscope which uses a single convex lens.



→ Magnification / magnifying power
~~Compound~~ Magnification is of compound microscope is higher than that of simple microscope as net magnification is equal to product of magnification of each lens used in compound microscope.

$$m = m_1 \times m_2 \times \dots m_n$$





→ For a small object placed near the objective lens (focal length = f_o), the lens create a real inverted magnified image of object between optic centre and focus of eye piece.

⇒ The eye piece consider this real image to be its virtual object and creates a virtual magnified image at least distance of distinct vision (at near point setting) magnified and

⇒ The final image is inverted with respect to object.

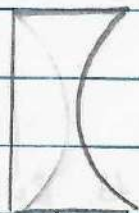
⇒ L denotes tube length of compound microscope, which is the distance between first second focus of objective lens and first focus of eye piece.

$$\text{magnifying power (at near point)} = \frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$$

(ii) Lens maker's formula,



$$\frac{1}{f} = (n_2 - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$



For thin plano concave lens,

$$P_1 = \frac{1}{f_1} = (n_1 - 1) \left(\frac{1}{\infty} - \frac{1}{R} \right)$$

$$= - \frac{(n_1 - 1)}{R}$$



For equiconvex lens,

$$P_2 = \frac{1}{f_2} = (n_2 - 1) \left(\frac{1}{R} + \frac{1}{R} \right) = \frac{2(n_2 - 1)}{R}$$

$$P_{\text{net}} = P_1 + P_2 = \frac{1}{f_1} + \frac{1}{f_2} = - \frac{(n_1 - 1)}{R} + \frac{2(n_2 - 1)}{R}$$

$$= \frac{2n_2 - n_1 - 1}{R} D //$$

31. (B)

(I).

$$j = \sigma E$$

$j \rightarrow$ current density.

$\sigma \rightarrow$ conductivity.

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Question Number

$$\frac{E_A}{E_B} = \frac{I_A}{I_B} = \frac{I}{A_A} \times \frac{A_B}{I} = \frac{2 \times 10^{-7}}{1 \times 10^{-7}} = 2.$$

(II) where $I = neA(v_d)$

$$v_d = \frac{I}{neA} = \frac{1}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 2 \times 10^{-7}}$$

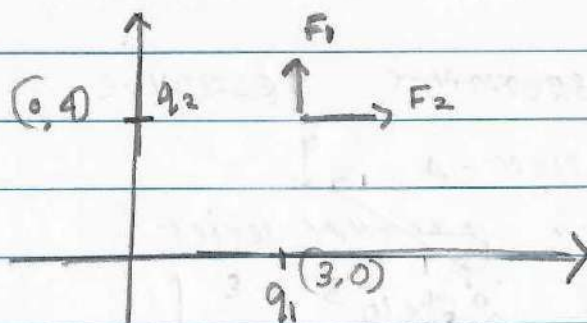
$$= \frac{10^{-28+27}}{8.5 \times 2 \times 16}$$

$$= \frac{10^{-1} \text{ m/s}}{17 \times 16}$$

$$= 0.36 \times 10^{-1} \text{ m/s}$$

$$= 3.6 \times 10^{-2} \text{ m/s}$$

(11).





$$E_{\text{net}} = \sqrt{E_1^2 + E_2^2}$$

$$= \sqrt{\left(\frac{kq_1}{(3)^2}\right)^2 + \left(\frac{kq_2}{(4)^2}\right)^2}$$

$$= k \sqrt{\frac{(16)^2}{9 \times 9} + \frac{1}{16 \times 16}} \times 10^{-6}$$

$$= \frac{k}{16 \times 9} \sqrt{(16)^4 + (3)^4} \times 10^{-6}$$

$$= \frac{9 \times 10^9 \times 10^{-6}}{16 \times 9} \sqrt{65536 + 81}$$

$$= \frac{10^3}{16} \sqrt{65617}$$

$$= 62.5 \sqrt{6517} \text{ N/C}$$

~~SECTION - C~~ SECTION - C

30

VB)

SECTION - A.

19.

Adding resistance in ~~parallel~~ series

$$V = IR \quad I = \frac{V}{R} = 25 \times 10^{-3}$$

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$$R = \frac{V}{I} - G$$
$$= 9000 \Omega$$

